



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

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CALIBRATION

Valid To: September 30, 2018

Certificate Number: 2737.03

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Electrical – DC & Low Frequency

Parameter/Equipment	Range	CMC ^{2, 4, 5} (±)	Comments
DC Voltage ³ – Generate	(0 to 330) mV (0.33 to 3.3) V (3.3 to 33) V (33 to 330) V (330 to 1000) V	23 μ V/V + 1 μ V 14 μ V/V + 2 μ V 14 μ V/V + 20 μ V 22 μ V/V + 0.15 mV 22 μ V/V + 1.5 mV	Fluke 5520A
DC Voltage ³ – Measure	(0 to 100) mV (0.1 to 1) V (1 to 10) V (10 to 100) V (100 to 1000) V	12 μ V/V + 0.3 μ V 11 μ V/V + 3 μ V 11 μ V/V + 0.5 μ V 13 μ V/V + 30 μ V 13 μ V/V + 100 μ V	HP 3458A
DC Current ³ – Generate	(0 to 3.3) mA (3.3 to 330) mA (0.33 to 3.3) A (3.3 to 20) A	0.17 mA/A + 50 nA 0.12 mA/A + 2.5 μ A 4.6 mA/A + 40 μ A 9.3 mA/A + 1.5 mA	Fluke 5520A

Parameter/Equipment	Range	CMC ^{2, 4, 5} ($\pm$)	Comments
DC Current ³ – Measure	(0 to 100) η A (0.1 to 1) μ A (1 to 10) μ A (10 to 100) μ A (0.1 to 1) mA (1 to 10) mA (10 to 100) mA (0.1 to 1) A	0.21 mA/A + 0.04 η A 59 μ A/A + 0.04 η A 27 μ A/A + 0.1 η A 26 μ A/A + 0.8 η A 26 μ A/A + 5 pA 26 μ A/A + 50 pA 46 μ A/A + 0.5 μ A 0.18 μ A/A + 10 μ A	HP 3458A
DC Resistance ³ – Generate	(0 to 10.999) Ω (11 to 32.999) Ω (33 to 109.999) Ω (110 to 329.999) Ω (0.33 to 1.0999) k Ω (1.1 to 3.2999) k Ω (3.3 to 109.999) k Ω (110 to 329.999) k Ω (0.33 to 1.0999) M Ω (1.1 to 3.2999) M Ω (3.3 to 10.999) M Ω (11 to 32.999) M Ω (33 to 109.999) M Ω (110 to 329.999) M Ω (330 to 1100) M Ω	55 $\mu\Omega/\Omega$ + 1 m Ω 42 $\mu\Omega/\Omega$ + 1.5 m Ω 94 $\mu\Omega/\Omega$ + 1.4 m Ω 34 $\mu\Omega/\Omega$ + 2 m Ω 34 $\mu\Omega/\Omega$ + 2 m Ω 37 $\mu\Omega/\Omega$ + 20 m Ω 19 $\mu\Omega/\Omega$ + 20 m Ω 37 $\mu\Omega/\Omega$ + 200 m Ω 37 $\mu\Omega/\Omega$ + 200 m Ω 40 $\mu\Omega/\Omega$ + 2 Ω 39 $\mu\Omega/\Omega$ + 2 Ω 73 $\mu\Omega/\Omega$ + 30 Ω 51 $\mu\Omega/\Omega$ + 50 Ω 0.32 m Ω/Ω + 2.5 k Ω 0.57 m Ω/Ω + 3 k Ω	Fluke 5520A
DC Resistance ³ – Measure	(0 to 10) Ω (10 to 100) Ω (0.1 to 1) k Ω (1 to 10) k Ω (10 to 100) k Ω (0.1 to 1) M Ω (1 to 10) M Ω (10 to 100) M Ω (0.1 to 1) G Ω	49 $\mu\Omega/\Omega$ + 50 $\mu\Omega$ 17 $\mu\Omega/\Omega$ + 500 $\mu\Omega$ 15 $\mu\Omega/\Omega$ + 500 $\mu\Omega$ 14 $\mu\Omega/\Omega$ + 500 $\mu\Omega$ 14 $\mu\Omega/\Omega$ + 5.0 $\mu\Omega$ 20 $\mu\Omega/\Omega$ + 50 $\mu\Omega$ 66 $\mu\Omega/\Omega$ + 1.0 Ω 3.4 m Ω/Ω + 100 Ω 6.5 m Ω/Ω + 10 k Ω	HP 3458A

Parameter/Equipment	Range	CMC ² (±)	Comments
Electrical Calibration of Thermocouples and Indicators ³ –			
Type B	(600 to 800) °C (800 to 1000) °C (1000 to 1550) °C (1550 to 1820) °C	0.35 °C 0.27 °C 0.24 °C 0.26 °C	Fluke 5520A
Type C	(0 to 150) °C (150 to 650) °C (650 to 1000) °C (1000 to 1800) °C (1800 to 2316) °C	0.24 °C 0.21 °C 0.25 °C 0.39 °C 0.65 °C	
Type E	(-250 to -100) °C (-100 to -25) °C (-25 to 350) °C (350 to 650) °C (650 to 1000) °C	0.53 °C 0.14 °C 0.12 °C 0.14 °C 0.17 °C	
Type J	(-210 to -100) °C (-100 to -30) °C (-30 to 150) °C (150 to 760) °C (760 to 1200) °C	0.22 °C 0.14 °C 0.12 °C 0.14 °C 0.19 °C	
Type K	(-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 1000) °C (1000 to 1372) °C	0.26 °C 0.15 °C 0.13 °C 0.21 °C 0.33 °C	
Type N	(-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 410) °C (410 to 1300) °C	0.31 °C 0.18 °C 0.16 °C 0.15 °C 0.22 °C	

Parameter/Equipment	Range	CMC ^{2, 4, 5} (±)	Comments
Electrical Calibration of Thermocouples and Indicators ³ – (cont)			
Type R	(0 to 250) °C (250 to 400) °C (400 to 1000) °C (1000 to 1767) °C	0.45 °C 0.28 °C 0.26 °C 0.31 °C	Fluke 5520A
Type S	(0 to 250) °C (250 to 1000) °C (1000 to 1400) °C (1400 to 1767) °C	0.37 °C 0.28 °C 0.29 °C 0.36 °C	
Type T	(-250 to -150) °C (-150 to 0) °C (0 to 120) °C (120 to 400) °C	0.49 °C 0.19 °C 0.14 °C 0.15 °C	
Type U	(-200 to 0) °C (0 to 600) °C	0.44 °C 0.22 °C	
Electrical Calibration of RTD Indicators ³ –			
Pt 385, 100 Ω	(-200 to 0) °C (0 to 100) °C (100 to 300) °C (300 to 400) °C (400 to 630) °C (630 to 800) °C	0.068 °C 0.055 °C 0.07 °C 0.098 °C 0.11 °C 0.18 °C	Fluke 5520A
Pt 3926, 100 Ω	(-200 to 0) °C (0 to 100) °C (100 to 300) °C (300 to 400) °C (400 to 630) °C	0.039 °C 0.055 °C 0.07 °C 0.078 °C 0.093 °C	
Pt 3916, 100 Ω	(-200 to -190) °C (-190 to -80) °C (-80 to 0) °C (0 to 100) °C (100 to 260) °C (260 to 300) °C (300 to 400) °C (400 to 600) °C (600 to 630) °C	0.2 °C 0.032 °C 0.039 °C 0.076 °C 0.073 °C 0.084 °C 0.09 °C 0.096 °C 0.18 °C	

Parameter/Equipment	Range	CMC ² (±)	Comments
Electrical Calibration of RTD Indicators ³ – (cont)			
Pt 385, 200 Ω	(-200 to -80) °C (-80 to 100) °C (100 to 260) °C (260 to 300) °C (300 to 400) °C (400 to 600) °C (600 to 630) °C	0.032 °C 0.032 °C 0.039 °C 0.093 °C 0.11 °C 0.11 °C 0.12 °C	Fluke 5520A
Pt 385, 500 Ω	(-200 to -80) °C (-80 to 100) °C (100 to 260) °C (260 to 400) °C (400 to 600) °C (600 to 630) °C	0.032 °C 0.039 °C 0.047 °C 0.062 °C 0.07 °C 0.086 °C	
Pt 385, 1000 Ω	(-200 to 0) °C (0 to 100) °C (100 to 260) °C	0.024 °C 0.032 °C 0.039 °C	
PtNi 385, 120 Ω	(-80 to 0) °C (0 to 100) °C (100 to 260) °C	0.062 °C 0.062 °C 0.11 °C	
Cu 427. 10 Ω	(-100 to 260) °C	0.23 °C	
Oscilloscope ³ –			
50 Ω Load	(1 to 24.999) mV (0.025 to 2.1999) V (2.2 to 130) V	0.25 % of output + 40μV 0.24 % of output + 40μV 0.27 % of output + 40μV	Fluke 5520A-SC1100
1 MΩ Load	(1 to 24.999) mV (25 to 109.999) mV (0.11 to 2.1999) V (2.2 to 10.999) V (11 to 130) V	0.12 % of output + 40μV 0.09 % of output + 40μV 0.31 % of output + 40μV 0.21 % of output + 40μV 0.14 % of output + 40μV	

Parameter/Equipment	Range	CMC ^{2, 4, 6} (±)	Comments
Oscilloscope ³ – (cont)			
Level Sine Wave:			
Amplitude (50kHz Reference)	50 kHz 50 kHz to 100 MHz (100 to 300) MHz (300 to 600) MHz (600 to 1100) MHz	1.6 % + 0.3 mV 2.8 % + 0.3 mV 3.2 % + 0.3 mV 4.7 % + 0.3 mV 5.7 % + 0.3 mV	Fluke 5520A- SC1100
Flatness (50kHz Reference)	50 kHz to 100 MHz (100 to 300) MHz (300 to 600) MHz (600 to 1100) MHz	1.4 % + 0.1 mV 1.6 % + 0.1 mV 3.2 % + 0.1 mV 3.9 % + 0.1 mV	
Time Markers: Source and Period into a 50 Ω load	5 s to 50 ms 20 ms to 2 ns	160 μs + t parts in 10 ⁶ s 64 μs + t parts in 10 ⁶ s	t = time in seconds
Amplitude 1 MΩ 50 Ω	1.8 mV to 55 V _(pk-pk) 1.8 mV to 2.5 V _(pk-pk)	2.4 % + 0.1 mV 2.4 % + 0.1 mV	
Frequency	10 Hz to 100 kHz	34 parts in 10 ⁶ Hz +15 mHz	

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Power ³ @ (45 to 65) Hz – (PF =1)			
(3.3 to 9) mA	(33 to 330) mV (0.33 to 1020) V	0.11 % 0.1 %	Fluke 5520A
(9 to 33) mA	(33 to 330) mV (0.33 to 1020) V	0.08 % 0.06 %	
(33 to 90) mA	(33 to 330) mV (0.33 to 1020) V	0.11 % 0.09 %	
(90 to 330) mA	(33 to 330) mV (0.33 to 1020) V	0.08 % 0.06 %	
(0.33 to 0.9) A	(33 to 330) mV (0.33 to 1020) V	0.1 % 0.1 %	

Parameter/Range	Frequency	CMC ^{2, 4, 6} ($\pm$)	Comments
AC Power ³ @ (45 to 65) Hz – (PF =1) (cont)			
(0.9 to 2.2) A	(33 to 330) mV (0.33 to 1020) V	0.09 % 0.07 %	Fluke 5520A
(2.2 to 4.5) A	(33 to 330) mV (0.33 to 1020) V	0.1 % 0.1 %	
(4.5 to 20.5) A	(33 to 330) mV (0.33 to 1020) V	0.09 % 0.31 %	
AC Voltage ³ – Generate			
(1 to 33) mV	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	0.064 % + 6 μ V 0.012 % + 6 μ V 0.016 % + 6 μ V 0.078 % + 6 μ V 0.27 % + 6 μ V 0.62 % + 6 μ V	Fluke 5520A
(33 to 330) mV	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	0.023 % + 8 μ V 0.011 % + 8 μ V 0.012 % + 8 μ V 0.027 % + 8 μ V 0.062 % + 32 μ V 0.16 % + 70 μ V	
(0.33 to 3.3) V	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	0.023 % + 50 μ V 0.012 % + 60 μ V 0.018 % + 60 μ V 0.027 % + 50 μ V 0.059 % + 125 μ V 0.22 % + 600 μ V	
(3.3 to 33) V	45 Hz to 1 kHz (1 to 10) kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.54 % + 0.65mV 0.026 % + 0.6 mV 0.11 % + 0.6 mV 0.15 % + 0.6 mV 0.17 % + 1.6 mV	
(33 to 330) V	45 Hz to 1 kHz (1 to 10) kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.23 % + 2 mV 0.23 % + 6 mV 0.24 % + 6 mV 0.40 % + 6 mV 0.43 % + 6 mV	

Parameter/Range	Frequency	CMC ^{2, 4, 5} ($\pm$)	Comments
AC Voltage ³ – Generate (cont) (330 to 1020) V	45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.41% + 10 mV 0.41% + 10 mV 0.40 % + 10 mV	Fluke 5520A
AC Voltage – Measure (0 to 10) mV (10 to 100) mV (0.1 to 1) V (1 to 10) V (10 to 100) V (100 to 10000) V	(1 to 40) Hz (40 to 1000) Hz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (10 to 40) Hz (40 to 1000) Hz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (1 to 40) Hz (40 to 1000) Hz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (1 to 40) Hz (40 to 1000) Hz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (1 to 40) Hz 40 Hz tto 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (1 to 40) Hz 40 Hz tto 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.039 % + 3 μ V 0.023 % + 1.1 μ V 0.034 % + 1.1 μ V 0.11 % + 1.1 μ V 0.57 % + 1.1 μ V 4.5 % + 2 μ V 0.049 % + 4 μ V 0.12 % + 2 μ V 0.045 % + 2 μ V 0.091% + 2 μ V 0.12 % + 2 μ V 0.35 % + 10 μ V 0.008 % + 40 μ V 0.008 % + 20 μ V 0.016 % + 20 μ V 0.034 % + 20 μ V 0.091 % + 20 μ V 0.34 % + 100 μ V 0.043 % + 400 μ V 0.027 % + 200 μ V 0.023 % + 200 μ V 0.041 % + 200 μ V 0.094 % + 200 μ V 0.34 % + 1 mV 0.21 % + 4 mV 0.21 % + 2 mV 0.21 % + 2 mV 0.28 % + 2 mV 0.31 % + 2 mV 1.6 % + 40 mV 1.7 % + 20 mV 2.3 % + 20 mV 2.3 % + 20 mV 0.36 % + 20 mV	Agilent 3458A-002

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
Capacitance ³ – Generate (0.19 to 0.3999) nF (0.4 to 1.0999) nF (1.1 to 3.2999) nF (3.3 to 10.9999) nF (11 to 32.9999) nF (33 to 109.999) nF (110 to 329.999) nF (0.33 to 1.0999) µF (1.1 to 3.29999) µF (3.3 to 10.9999) µF (11 to 32.9999) µF (33 to 109.999) µF (110 to 329.999) µF (0.33 to 1.09999) mF (1.1 to 3.2999) mF (3.3 to 10.9999) mF (11 to 32.9999) mF (33 to 110) mF	10 Hz to 10 kHz 10 Hz to 10 kHz 10 Hz to 3 kHz 10 Hz to 1 kHz 10 Hz to 1 kHz 10 Hz to 1 kHz 10 Hz to 1 kHz (10 to 600) Hz (10 to 300) Hz (10 to 150) Hz (10 to 120) Hz (10 to 80) Hz (10 to 50) Hz (10 to 20) Hz (0 to 6) Hz (0 to 2) Hz (0 to 0.6) Hz (0 to 0.2) Hz	0.55 % + 0.01 nF 0.42 % + 0.01 nF 0.43 % + 0.01 nF 0.19 % + 0.01 nF 0.19 % + 0.1 nF 0.19 % + 0.1 nF 0.19 % + 0.3 nF 0.19 % + 1 nF 0.19 % + 3 nF 0.19 % + 10 nF 0.31 % + 30 nF 0.36 % + 100 nF 0.35 % + 300 nF 0.35 % + 1 µF 0.35 % + 3 µF 0.35 % + 10 µF 0.58 % + 30 µF 0.85 % + 100 µF	Fluke 5520A
AC Current ³ – Generate (29 to 329.99) µA (0.33 to 3.2999) mA	(10 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz (10 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.16 % + 100 nA 0.097 % + 100 nA 0.23 % + 150 nA 0.62 % + 200 nA 1.2 % + 400 nA 0.16 % + 150 nA 0.078 % + 150 nA 0.16 % + 200 nA 0.39 % + 300 nA 0.78 % + 600 nA	Fluke 5520A

Parameter/Range	Frequency	CMC ^{2, 4, 5, 6} (±)	Comments
AC Current ³ – Generate (cont)			
(3.3 to 32.999) mA	(10 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.14 % + 2 µA 0.10 % + 2 µA 0.062 % + 2 µA 0.16 % + 3 µA 1.1 % + 4 µA	Fluke 5520A
(33 to 329.99) mA	(10 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.14 % + 20 µA 0.032 % + 20 µA 0.078 % + 50 µA 0.16 % + 100 µA 0.31 % + 200 µA	
(0.33 to 1.09999) A	(10 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.14 % + 100 µA 0.04 % + 100 µA 0.47 % + 1 mA 1.9 % + 5 mA	
(1.1 to 2.99999) A	(10 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.21 % + 100 µA 0.13 % + 100 µA 0.14 % + 1 mA 2 % + 5 mA	
(3 to 10.9999) A	(45 to 100) Hz (0.100 to 1) kHz (1 to 5) kHz	0.31 % + 2 mA 0.31 % + 2 mA 2.4 % + 2 mA	
(11 to 20.5) A	(45 to 100) Hz 100 Hz to 1 kHz (1 to 5) kHz	0.12 % + 5 mA 0.15 % + 5 mA 3 % + 5 mA	
AC Current ³ – Measure			
Up to 100 µA (0.1 to 1) mA (1 to 10) mA (10 to 100) mA (0.1 to 1) A	40 Hz to 5 kHz	0.071 % + 31 nA 0.036 % + 0.21 µA 0.036 % + 2.1 µA 0.036 % + 21 µA 0.12 % + 0.21 mA	HP 3458A-002

II. Electrical – RF/Microwave

Parameter/Range	Frequency	CMC ^{2, 6} (±)	Comments
RF Absolute Power – Measure			
Power Reference ³			Weinschel 1805/1807
1 mW, Type-N(f), 50Ω	50 MHz	1.2 %	
Coaxial Power Sensors	10 MHz to 6 GHz	2.6 %	

III. Mechanical

Parameter/Equipment	Range	CMC ^{2, 6} (±)	Comments
Acceleration – Measure	20 Hz to 10 kHz	2.6 %	Dytran 3120B reference accelerometer

IV. Time & Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
Time Interval	Up to 86 400 s	30 ms	Agilent 53131A-030 counter phase locked to GPS
Frequency – Measure	1 mHz to 40 GHz	$1.2 \times 10^{-9} \times f + 5 \mu\text{Hz}$	Agilent 53131A-030, HP 5352B counter phase locked to GPS (<i>f</i> is the frequency)
Frequency – Measuring Equipment	10 MHz 1 mHz to 50 GHz	$1.2 \times 10^{-9} \times f$ $1.2 \times 10^{-9} \times f + 5 \mu\text{Hz}$	Datum GPS receiver Agilent 3325B Agilent 83650B Phase locked to GPS (<i>f</i> is the frequency)

¹ This laboratory offers commercial calibration and field calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴ The measurements stated are generated with the Fluke 5500 series of instruments. This capability is suitable for the calibration of the devices intended to measure the stated measurement in the ranges indicated. Best measurement uncertainties are expressed as either a specific value that covers the full range or as a fraction of the reading plus a fixed floor specification.

⁵ The measurements stated are measured with the HP 3400 series of instruments. This capability is suitable for the calibration of the devices intended to generate the measurement in the ranges indicated. Best measurement uncertainties are expressed as either a specific value that covers the full range or as a combination of the fraction of the reading/output plus a range specification.

⁶ In the statement of CMC uncertainty, the value is defined as the percentage of reading unless otherwise indicated.

⁷ L represents the length of displacement in inches. R represents the resolution of the unit under test.



Accredited Laboratory

A2LA has accredited

ACUCAL INC.

Chandler, AZ

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSLI Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 19th day of August 2016.

A handwritten signature in blue ink, appearing to read 'J. C. Bunt'.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 2737.03
Valid to September 30, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.